

User Guide: ZIP Code 3.0

Overview This application processes ZIP code data from a CSV or structured file format. It allows users to:

- Convert standard CSV files into a structured, length-indicated format
- Generate summaries of geographic extremes (east, west, north, south) by state
- Search and display ZIP code records using command-line flags

Launching the Application

Compilation To compile the application: `make`

Running You can run the program in two modes:

1. Interactive Menu Mode (default) `./myProgram` This prompts the user to:
 - Convert a CSV to a structured format
 - Process a file to generate a summary
 2. ZIP Code Search Mode `./myProgram ZipCodes.txt -z56301 -z90210 -z99999`
 - ZipCodes.txt is the length-indicated data file
 - Each `-z#####` is a ZIP code to search for
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File Formats

Input CSV The CSV must have the following header: "Zip Code","Place Name","State","County","Latitude","Longitude"

Length-Indicated Format This format includes:

- Header metadata
- Records prefixed by their length

It is automatically generated using option 1 in the interactive mode.

Features

Convert CSV to Structured Format

- Launch the program
- Choose option 1
- Provide input and output filenames

Generate State Summary

- Choose option 2
- Select a sort field: zipCode, placeName, state, latitude, or longitude
- Output is written to a CSV file (name provided by the user)

Lookup ZIP Codes (Command-line)

- Run with `-z#####` flags
- Matching records are printed
- If a ZIP code isn't found, a message is shown

Sample Output

State Summary Output State, Easternmost, Westernmost, Northernmost, Southernmost
NY, 10001, 14905, 10598, 10002 CA, 90210, 96162, 96162, 90001

ZIP Search Output Zip Code: 56301, Place Name: Saint Cloud, State: MN, County: Stearns,
Latitude: 45.555, Longitude: -94.167 Zip Code 99999 not found in the file.

Troubleshooting • "Could not open file": Check the file name and path (case-sensitive) •
Malformed data: Ensure your CSV uses comma separators and proper quoting • Missing
header: First row of CSV must match expected field names

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